

API Testing: From Zero to Confident in 30 Days

The Exact Roadmap I Wish I Had

- **Week 1** Fundamentals
- **Week 2** Hands-On
- **Week 3** Going Deeper
- **Week 4** Real-World

Aston Cook
Senior QA Automation Engineer

WEEK 1

The Fundamentals

DAY 1-2

Learn what an API actually is. Understand REST, endpoints, and the request/response cycle. Don't touch a tool yet.

DAY 3-4

Learn HTTP methods: GET, POST, PUT, DELETE, PATCH. Understand when each is used and why.

DAY 5-6

Learn status codes. Not just 200 and 404. Learn 201, 204, 400, 401, 403, 409, and 500.

DAY 7

Set up Postman or your preferred tool. Send your first requests to a public API.

Aston Cook

Senior QA Automation Engineer

WEEK 1

Pro Tip

“

Don't rush past the fundamentals. I've seen engineers with 2 years of experience who can't explain the difference between PUT and PATCH. Understanding the 'why' behind each method makes everything else click faster.

Aston Cook

Senior QA Automation Engineer

WEEK 2

Hands-On Practice

DAY 8-9

Test GET endpoints. Validate response body, headers, and status codes.

DAY 10-11

Test POST endpoints. Send request bodies, validate created resources.

DAY 12-13

Learn authentication. Practice with API keys, Bearer tokens, and OAuth basics.

DAY 14

Chain requests together. Use the response from one call as input for the next.

Aston Cook

Senior QA Automation Engineer

WEEK 2

Pro Tip

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Use a free public API to practice. JSONPlaceholder, ReqRes, or the PokeAPI are great starting points. Don't wait until you have a 'real' project to start testing.

Aston Cook

Senior QA Automation Engineer

WEEK 3

Going Deeper

DAY 15-16

Learn response schema validation. Verify data types, required fields, and structure.

DAY 17-18

Test negative scenarios. Invalid inputs, missing fields, wrong data types, unauthorized access.

DAY 19-20

Learn about headers in depth. Content-Type, Authorization, and rate limiting headers.

DAY 21

Write your first automated API test using Playwright or your preferred framework.

Aston Cook

Senior QA Automation Engineer

WEEK 3

Pro Tip

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Negative testing is where you find the real bugs. Any tool can verify a happy path. The engineers who stand out are the ones who break things on purpose and find what the developers missed.

Aston Cook

Senior QA Automation Engineer

WEEK 4

Real-World Skills

DAY 22-23

Test with dynamic data. Environment variables, data-driven tests, parameterization.

DAY 24-25

Learn about API mocking and stubbing. When and why to mock responses.

DAY 26-27

Integrate API tests into a CI/CD pipeline. Automate the run.

DAY 28-29

Build a mini test suite with reporting. Cover CRUD operations end to end.

DAY 30

Review, refactor, and document your test suite. You made it.

Aston Cook

Senior QA Automation Engineer

WEEK 4

Pro Tip

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Your API test suite should run in under 2 minutes. If it takes longer, you're probably testing too much through the API that should be covered by unit tests. Keep it lean and fast.

Aston Cook

Senior QA Automation Engineer

WATCH OUT

5 Mistakes to Avoid

1

Only testing happy paths

Test the edge cases and error scenarios too

2

Hardcoding test data

Use variables, factories, and env configs

3

Ignoring response time

Slow APIs mean bad user experience

4

Skipping auth tests

Security bugs are the most expensive to fix

5

Not validating schemas

Structure matters as much as the data itself

Aston Cook

Senior QA Automation Engineer

10 / 12

AFTER 30 DAYS

You Will Be Able To

- ✓ Confidently test any REST API endpoint
- ✓ Validate schemas, headers, and edge cases
- ✓ Write automated API tests in code
- ✓ Integrate tests into CI/CD pipelines
- ✓ Talk about API testing in interviews without freezing

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Senior QA Automation Engineer

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30 Days. That's All It Takes to Go From Zero to Confident.

API Mastery for QA

Go deeper with hands-on projects. Available now.

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Aston Cook

Senior QA Automation Engineer